

Intelligent Workflow Automation – High-Level Technical Specification

Objective

Build a modern SaaS platform that enables business users to configure and test automated workflows between systems.

The platform should focus on simplifying integration setup, data transformation, validation, execution monitoring, and operational visibility.

Core Capabilities

1. Workflow Configuration

Users should be able to:

- Create and manage workflow definitions.
 - Define source and destination systems.
 - Configure execution triggers.(Schedules)
 - Configure field mappings.
 - Define **validation** and transformation **rules**. (OrderId, Order Number etc)
 - Save and version workflow configurations.
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2. Data Mapping Engine

Provide a visual experience that allows users to:

- Map source fields to destination fields.
- Support **nested** structures.
- Apply simple transformations.
- Define **default** values.
- Support conditional logic.(natural language)

- Validate mappings before execution.

Examples of supported transforms:

- Trim text
 - Case conversion
 - Type conversion
 - Default value assignment
 - Basic conditional expressions
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3. Rule Engine

Implement a configurable rules framework supporting:

- Conditional evaluation.
- AND/OR groupings.
- Validation rules.
- Simulation/testing of rules.

Rules should be configurable by users through UI configuration rather than code.

4. Connector Framework

The platform should expose a pluggable connector architecture supporting:

- REST APIs(Authentication support,Pagination support)
- Webhooks(think about it*)
- File-based integrations(xml, csv, txt)
- Databases (temp table)
- Future connector extensibility(List of connectors)

5. Test & Simulation Module

Users should be able to:

- Execute workflows in test mode.
- Simulate external responses.

- Run multiple test scenarios.
- Review execution traces.
- Compare expected vs actual outcomes.

Test results should include:

- Success/failure status
 - Validation errors
 - Processing duration
 - Detailed execution logs
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7. Monitoring & Operations

Provide operational dashboards showing:

- Workflow health
- Success/failure rates
- Throughput metrics
- Recent executions
- Error summaries
- Execution history

Operational capabilities:

- Search runs
 - Reprocess failed executions
 - View execution timeline
 - Export diagnostics
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8. Error Management

Platform should provide:

- Centralized exception handling.
- Error categorization.
- Human-readable error messages.
- Correlation identifiers.
- Retry recommendations.

- Failed record management.

Technical Expectations

The ideal solution should demonstrate:

- Clean Architecture / Domain-Driven Design.
- Modular architecture.
- Extensibility through interfaces/plugins.
- Testability.
- Strong separation of concerns.
- Scalable asynchronous processing.
- Cloud-ready deployment model.

Preferred technology stack:

- .NET / ASP.NET Core
- C#
- SQL database
- Background job processing
- Container-ready deployment